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A Model Summary

A summary of the models examined and utilized in evaluations and case studies is presented in Table 2. We selected these models to encompass a broad range of capabilities, spanning both open-source and closed-source models. We also consider different overlaps between release and knowledge cut-off dates so that the identifier models are evaluated on target models with release dates both before and after the knowledge cut-off dates. The models are plotted in Figure 8 to show the overlaps between models' training cut-off dates and release dates. For closed-source models, we use the API call directly from the providers. For open-source models, we use the API call from Together AI (Together AI, n.d.).

Weight Type	Model	Platform (Provider)	Released	Training cut-off	Parameters
Open source	Mistral Instruct v0.3 (Jiang et al., 2023)	Together (Mistral)	May 2024	Unknown	7B
	Llama 3.1 (Grattafiori et al., 2024)	Together (Meta)	Jul 2024	Dec 2023	8B
	Llama 3.3 (Meta Platforms, 2024)	Together (Meta)	Dec 2024	Dec 2023	70B
	Deepseek R1 (DeepSeek-AI, 2025)	Deepseek	Jan 2025	Jul 2024	671B
	Deepseek V3 (DeepSeek-AI, 2024)	Deepseek	Mar 2025	Jul 2024	671B
	Qwen-2.5 Instruct (Qwen et al., 2025)	Together (Alibaba)	Sep 2024	Oct 2023	72B
	Qwen-3 (Yang et al., 2025)	Together (Alibaba)	Apr 2025	Unknown	235B
Closed source	GPT-4o-mini (OpenAI, 2024b)	OpenAI	Jul 2024	Oct 2023	?
	GPT-o4-mini (OpenAI, 2025)	OpenAI	Apr 2025	Jun 2024	?
	Claude-3.5-Haiku (Anthropic, 2024)	Anthropic	Oct 2024	Jul 2024	?
	Claude-3.7-Sonnet (Anthropic, 2025)	Anthropic	Feb 2025	Oct 2024	?

Table 2: Overview of the models evaluated and utilized in case studies.

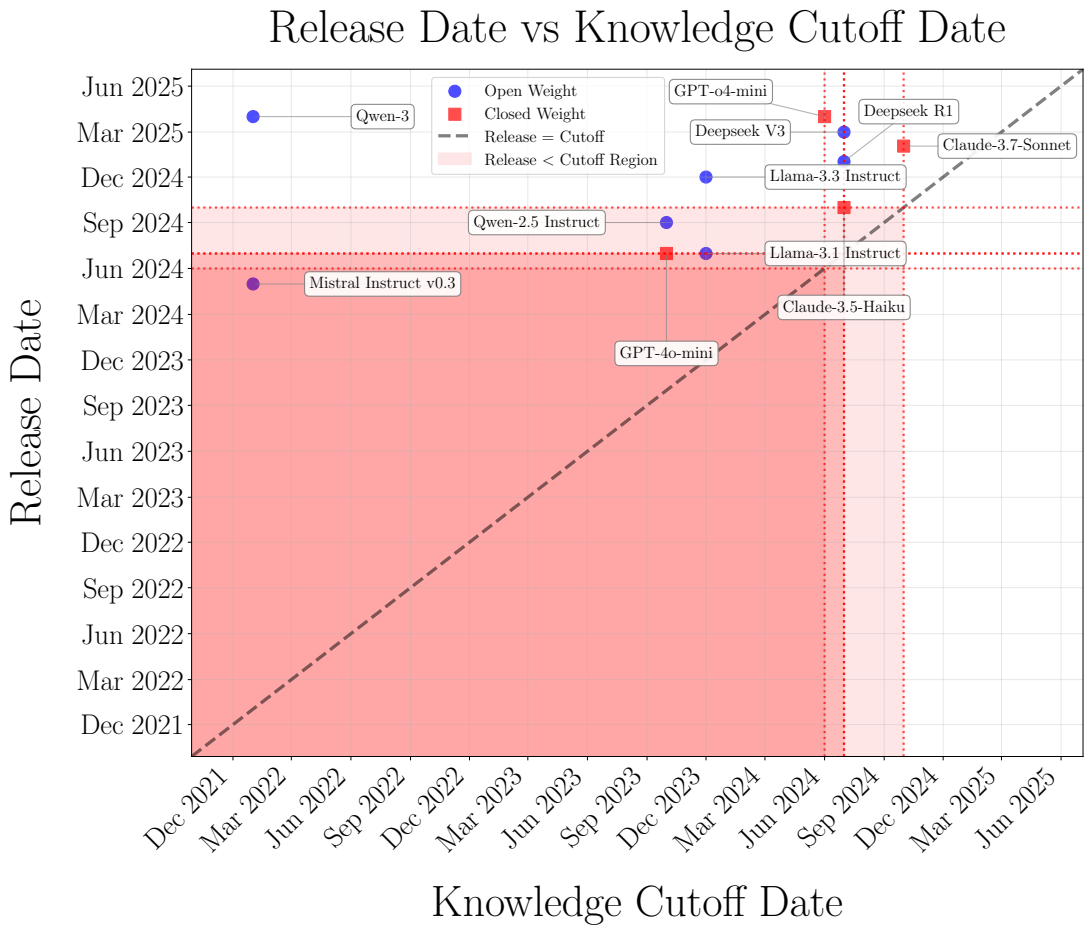


Figure 8: Overview of models' release date and knowledge cutoff date.

	Dataset	Domain	Info
Evaluation Reasoning	MATH (Hendrycks et al., 2021a)	Mathematics	Challenging problems from mathematics competitions such as AMC 10, AMC 12, and AIME. (We use level 5 problems for evaluation.)
	HumanEval (Chen et al., 2021b)	Coding	164 Python programming problems, where each problem includes a function signature, docstring, and unit tests.
Evaluation Linguistic Style	XSum (Narayan et al., 2018)	Summarization	226,711 news articles accompanied with a one-sentence summary for evaluation of abstractive single-document summarization.
	UltraChat (Ding et al., 2023)	Dialogue	Multi-round dialogue data consisting of world questions, creative writing, and writing recreation.
Evaluation Alignment Preference	Value Kaleidoscope (Sorensen et al., 2024b)	Human Value	218k values, rights, and duties connected to 31k human-written situations.
	Election Questions (Anthropic, 2024)	Politics	Evaluation dataset to assess language models' ability to handle election-related information accurately and harmlessly without engaging in persuasion targeting.
Case study datasets	MATH (Hendrycks et al., 2021a)	Mathematics	Challenging problems from mathematics competitions such as AMC 10, AMC 12, and AIME. (We use level 4 problems for the case study.)
	Chatbot Arena (Chiang et al., 2024)	Conversation	33 K cleaned conversations with pairwise human preferences.
	JailbreakBench (Chao et al., 2024)	Safety	An evolving dataset of state-of-the-art adversarial prompts consisting of 100 distinct misuse behaviors.

Table 3: Overview of evaluation and case study datasets.

B Dataset Details

The detailed information for the evaluation and case study datasets is listed in Table 3. For all experiments, we randomly sampled 100 data points from each respective dataset to manage computational costs.

C Details And Additional Results for Evaluation

We describe the detailed implementation of the evaluation and present the additional evaluation results in this section.

C.1 Implementation details

All model inferences and response generations throughout the experiments were conducted via the official APIs for each respective model. We utilized the default hyperparameter settings as specified by each provider (e.g., for temperature, top_p, and maximum tokens) to ensure our interactions reflected standard operational behavior. No model fine-tuning or modification of underlying weights was performed.

For the evaluation setup, after a target model generated a response to a task prompt, the identifier model was presented with this response. The identifier's task was to select the correct model family from a multiple-choice list, formatted as follows: (A) GPT (B) Claude (C) DeepSeek (D) Llama (E) Mistral (F) Cannot Identify. The inclusion of a "(F) Cannot Identify" option was designed to reduce forced-choice errors and allow the model to express uncertainty rather than generating a random guess.

To specifically probe how different aspects of a response contribute to identifiability, we used three distinct identification prompt templates. Each template explicitly instructed the identifier to focus its analysis on one of three dimensions: (1) reasoning patterns, (2) linguistic style, or (3) alignment preferences. The full text of these dimension-specific prompt templates can be found in Appendix C.2.